

REAL-WORLD DATA PROJECT

Retail Sales Data Analysis and Prediction

Domain: Retail

Project Type: End-to-End Data Analysis / Prediction

Prepared for: Academic Data Science Task

Project Objective

This project analyzes a real-world retail sales dataset to understand sales performance, identify useful business patterns, and build a simple prediction model. The project covers the complete data science workflow: dataset selection, data cleaning, exploratory analysis, visualization, model building, evaluation, and conclusions.

1. Problem Statement

Retail businesses generate large amounts of sales data. Studying this data can help identify high-performing products, understand customer purchasing patterns, compare sales across time periods, and support better business decisions. The aim of this project is to convert raw retail sales data into meaningful insights and use historical information to estimate future sales.

2. Dataset Description

A retail sales dataset is used for this project. Typical fields include transaction date, product or category, quantity sold, unit price, customer information, and total sales amount. The dataset can be obtained from a public dataset repository such as Kaggle or another open data source.

Suggested dataset: Retail Sales Dataset containing transaction-level sales records.

3. Tools and Technologies

Tool / Technology	Purpose
Python	Data analysis and model development
Pandas	Data cleaning and manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Scikit-learn	Machine learning and evaluation
Jupyter Notebook / Google Colab	Project development and execution

4. Data Preprocessing

The raw dataset is first inspected for missing values, duplicate records, incorrect data types, and inconsistent entries. Date columns are converted into date format. Numerical columns such as quantity, price, and sales are checked for invalid values. Duplicate rows are removed where necessary, and missing values are handled using an appropriate method.

5. Exploratory Data Analysis

Exploratory analysis is performed to understand the main characteristics of the retail data. Important measures include total sales, average sales, quantity sold, sales by product/category, and sales by month. Grouping and aggregation are used to identify trends and differences between products and time periods.

6. Visualizations

The project presents the findings using clear charts. Suggested visualizations include:

Visualization	Purpose
Monthly Sales Line Chart	Identify sales trends over time
Category-wise Bar Chart	Compare sales between categories
Top Products Chart	Find products with the highest sales
Quantity vs Sales Scatter Plot	Study the relationship between quantity and revenue

7. Prediction Approach

A simple machine learning model can be developed to predict sales. Historical transaction information is divided into training and testing data. Relevant numerical and time-based features are selected. A regression model such as Linear Regression or Random Forest Regression can then be trained on the training data and evaluated on unseen test data.

8. Model Evaluation

The prediction model can be evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score. These metrics help measure how close the predicted sales values are to the actual values. The final notebook should report the actual metric values obtained from the selected dataset.

9. Expected Findings

The analysis is expected to reveal differences in sales across products and categories, changes in sales over time, and relationships between quantity, price, and revenue. The exact findings should be reported from the executed analysis rather than assumed in advance.

10. Business Insights

The results can support retail decision-making by identifying products or categories that contribute strongly to revenue, recognizing periods of higher or lower demand, and providing a data-based estimate of sales. These insights can help with inventory planning, sales strategy, and resource allocation.

11. Project Workflow

Dataset Collection → Data Cleaning → Exploratory Data Analysis → Visualization → Feature Preparation → Model Training → Model Evaluation → Insights → Conclusion

12. Conclusion

This project demonstrates an end-to-end application of data science to a real-world retail problem. Raw sales data is transformed into useful information through preprocessing, analysis, visualization, and prediction. The project shows how data science techniques can be applied to understand business performance and support evidence-based decisions.

13. GitHub Repository Structure

File / Folder	Purpose
README.md	Project overview, objectives, dataset, methods, results, and conclusion
retail_sales_analysis.ipynb	Complete Python analysis and prediction notebook
data/	Dataset used for the project, subject to its license
images/	Charts and visualizations generated by the notebook
requirements.txt	Python libraries required to run the project
PROJECT_REPORT.pdf	This project documentation

Note: Before submission, replace generic dataset/model descriptions with the exact dataset name, row/column counts, charts, model metrics, and findings produced by the executed notebook.